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## **Home Value Prediction Analysis**

### **Abstract/Executive Summary**

This project investigates how accurately we can forecast median home values in U.S. metropolitan areas using historical data on home prices and sales activity. Specifically, we focused on predicting the Zillow Home Value Index (ZHVI) for New York, NY, one year into the future using the prior 12 months of ZHVI and home sales data. The primary objective was to evaluate the effectiveness of different modeling approaches in capturing time-dependent housing market trends.

We trained and compared three predictive models: LASSO regression, Random Forest Regressor, and XGBoost Regressor, using a time-series aware cross-validation approach to prevent data leakage. Our models were evaluated using two common error metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Surprisingly, the LASSO regression model outperformed the more complex machine learning models, achieving the lowest RMSE (29,834) and MAE (24,477). Despite being a linear model, LASSO's regularization likely helped it generalize well on this trend-dominated dataset. Tree-based models underperformed slightly: Random Forest yielded an RMSE of 45,844, and XGBoost improved upon that with an RMSE of 42,810.

Feature importance analysis revealed that recent ZHVI values (within 1–3 months) were far more predictive than older data or home sales counts, suggesting that home values are highly

momentum-driven. Home sales data contributed little to model accuracy, possibly due to its lagging nature or noise.

Overall, the findings suggest that simple, regularized models can provide surprisingly strong forecasting performance when data trends are stable. However, the exclusion of macroeconomic indicators limits the explanatory scope of our models. Future extensions could integrate broader economic data and evaluate generalizability across other cities.

This work offers a practical foundation for forecasting home prices and informs housing market stakeholders looking to anticipate short-term market movements using historical trends.

## **Introduction**

The prediction of home value is critical for both economic planning and investment strategies in real estate. With the rise of big data and advanced machine learning techniques, there is a growing opportunity to leverage historical data for forecasting home prices with increased accuracy. This study focuses on utilizing Zillow Home Value Index (ZHVI) and home sales count data to predict median home values in major U.S. metropolitan areas with particular emphasis on New York, NY.

Our primary objective is to explore whether historical trends in ZHVI and home sales activity can be predictive of future home values, one year ahead. Specifically, we will evaluate the effectiveness of different machine learning models, including LASSO regression, Random Forest, and XGBoost, to capture time dependent patterns in housing prices. These patterns are influenced by various market factors including economic shifts, buyer confidence and seasonal variations in home sales. We hypothesize that by modelling these trends, we can derive accurate short term forecasts for home values.

In this paper, we present a detailed analysis of the forecasting models and compare their performance using key metrics such as RMSE and MAE. We assess the relative importance of different features, such as recent ZHVI values and home sales counts, in predicting future home values. The results from this analysis are intended to guide future work in predictive modeling of real estate markets and provide actionable insights for real estate investors, policy makers, and other market stakeholders.

## **Data**

The dataset consists of two primary sources of real estate market information: the Zillow Home Value Index (ZHVI) and home sales count data, both provided by Zillow. The ZHVI dataset offers a time-series representation of median home values across hundreds of U.S. metropolitan areas, with monthly values dating back to at least 2008. The home sales dataset tracks the number of properties sold per month in these same metro areas. Each dataset includes common identifiers (RegionID, RegionName, and StateName) allowing for accurate merging across sources and consistent geographic alignment.

This project focuses on answering the question: Can historical home values and sales activity be used to predict median home values one year into the future in major U.S. cities, such as New York? By studying time-series patterns in ZHVI and sales counts, we aim to better understand how these signals evolve over time, and whether they hold predictive value for short-term market forecasting.

The ZHVI trends are especially useful in revealing long-term price appreciation, housing affordability changes, and responses to macroeconomic shifts. Meanwhile, fluctuations in sales counts may reflect changes in supply and demand dynamics, buyer confidence, and seasonal

variation. By combining both sources, we can explore whether sales trends precede or follow price movements, and whether recent activity levels help predict future valuations.

Before modeling, substantial data cleaning was required. This included dropping rows with missing region identifiers and aligning both datasets across a shared timeline. Additionally, we generated lag features from each time series to allow the models to “look back” 12 months when making forward predictions. Finally, we restricted our analysis to one representative city, New York, NY, for a focused case study with robust data availability.

## **Methods**

In this study, we aimed to investigate whether historical trends in home values and sales activity can predict median home values one year ahead in major U.S. metropolitan areas. To achieve this, we used supervised learning with regression techniques.

Our target variable is home sales count. It's a continuous variable which makes it suitable for supervised regression models. We began by using linear regression to establish a baseline model and understand the relationship between historical data and future home values. We chose LASSO regression to help with multicollinearity that came up when using lagged monthly values. The LASSO model adds regularization to the regression process, helping to reduce overfitting by penalizing overly complex models. We also implemented a Random Forest Regression model to capture potential non-linear relationships between variables. Random Forests are beneficial in capturing non-linear patterns in the data which are often found in real estate markets and can provide deeper insights into how different features interact.

We evaluated the performance of each model using two primary metrics: Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). These were used to assess the accuracy of predictions on the test set. Since our data has a time-series structure, we used a

time-aware train/test split where the model was trained on data from earlier years and tested on more recent years. This approach simulates real world forecasting scenarios and mitigates the risk of data leakage. Additionally, to determine whether our models provided a valuable prediction, we compared them against a simple baseline model which predicted next year's home value by carrying forward the previous year's value. This comparison allowed us to assess if the more complex models were actually better at predicting than just using a simple trend.

We expected a few challenges, like missing or inconsistent data and high dimensionality. To handle missing values, we used interpolation and took a closer look at areas with big gaps to decide if those points should be removed. Since we were working with time-series data, having too many variables could add noise. To address this issue, we thought about splitting the data into seasonal groups and using rolling averages to smooth out short-term changes. If our models didn't perform well, we figured it might be because the features or time periods we picked weren't strong enough to predict home values. In that case, we planned to shift our focus to longer-term trends, like yearly averages, to cut down on noise and bring out more useful patterns.

Next, we process the dataset by converting monthly time series into lagged features, computing yearly changes in ZHVI and sales counts, and incorporating rolling averages to reduce noise. To explore market similarities, we will also apply clustering techniques to group similar markets which may enhance the models ability to detect patterns. These steps, combined with regression metrics, model coefficients and visualizations, will help us present a comprehensive analysis of our predictive models.

## **Results**

We started by cleaning and merging the two primary datasets: the Zillow Home Value Index (ZHVI) and home sales count data. The initial cleaning involved removing rows with

missing 'StateName' values and aligning the datasets across a shared timeline (2008 onwards). The cleaned dataset retained geographical identifiers ('RegionID', 'RegionName', 'StateName') and included columns for both historical ZHVI values and home sales counts. After merging, the final dataset consisted of 415 columns representing monthly ZHVI and sales values from 2008-2024.

We calculated summary statistics to understand the distribution and variation of both ZHVI and sales data. The average ZHVI increased significantly from 2010 to 2024. In 2010, the mean ZHVI was approximately \$193,747, which increased to \$359,179 by 2021, revealing a consistent upward trend in home values. Home sales counts also varied significantly, with a visible trend in more recent years. For example, the average sales count rose from 1,685 in 2010 to 3,421 in 2021. The initial summary statistics highlighted home values are momentum-driven, with sales counts displaying positive correlation with ZHVI increases.

Following the initial summary statistics check, we visualized the ZHVI and home sales trends for major metro areas. The most notable results included home value trending upwards consistently, particularly in cities such as New York and Los Angeles. Correlation heatmaps for ZHVI revealed strong positive correlations among recent months, confirming the momentum-driven nature of home values. Additionally, sales count correlations were much weaker, displaying sales do not directly influence short-term values changes.

We trained three models: LASSO regression, Random Forest Regressor, and XGBoost Regressor. The models were evaluated using Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE). Lasso regression achieved the lowest RMSE (29,834) and MAE (24,277), outperforming the more complex models. The model was able to adequately reduce overfitting and capture linear trends due to its regularization. Random Forest Regressor's RMSE

of 45,844, displayed a poorer performance which we can account for overfitting due to noisy features. For XGBoost Regressor, with a RMSE of 42,810, we noticed that capturing non-linear relationships did not significantly improve forecasting accuracy. The most predictive features were recent ZHVI values, especially those from the past 1-3 months, highlighting a strong influence of recent trends on short-term forecasting. Alternatively, home sales data had minimal impact on model accuracy, due to its lagging natures and additional noise.

The results showcase simple, regularized models that can outperform more complex methods when dealing with trend-dominated datasets such as the ones we chose. The findings underscore the importance of recent price trends as key predictors and suggest limited utility in incorporating home sales counts for short-term value forecasting.

## **Conclusions**

Through this study, we attempted to predict median home values one year into the future by leveraging historical data from the Zillow Home Value Index (ZHVI) and home sales count. With a focus on major U.S. cities, such as New York and Los Angeles, we studied whether trends in housing prices and sales data could provide reliable forecasts. We ventured to study this through three machine learning models: LASSO regression, Random Forest and XGBoost. These models allowed us to evaluate the predictive power of these variables. The primary performance metrics used were Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

The LASSO regression model consistently outperformed the more complex tree-based models, achieving the lowest RMSE and MAE. This result highlights simpler, linear models that can effectively capture price trends when the underlying data exhibits stable, momentum-driven patterns. In contrast, Random Forest and XGBoost struggled to improve upon LASSO's

performance, likely due to overfitting and the inherent noise in home sales data. Our feature importance analysis confirmed that recent ZHVI values were more predictive than older data, while home sales had a negligible impact.

One significant challenge we encountered during this study was the variability and noise inherent in the sales data. While ZHVI trends demonstrate consistent upward patterns, sales counts fluctuated considerably, making them less reliable as predictors. Additionally, the lack of macroeconomic factors, such as interest rates, employment statistics or inflation data, may have limited the model's abilities to capture broader market dynamics.

Tree-based models like Random Forest and XGBoost struggled to predict sharp increases in home values, likely because they average across many decision trees, capturing general trends well but being less responsive to sudden jumps. On the other hand, LASSO, as a linear model, performed well for trend prediction but missed more complex nonlinear relationships, particularly in unpredictable price changes like the post-COVID housing boom.

Another limitation stemmed from the geographical focus on New York, NY. While this city offers robust data and a well-defined market trend, the findings may not generalize to cities with different housing market characteristics, such as those with more volatile economic conditions or less predictable real estate cycles.

We also faced challenges with the dataset itself. Some metro areas had missing monthly data or inconsistent region names, complicating the merging and alignment process. Since the data was at a monthly level, it also introduced significant short-term noise, making trend interpretation challenging. To address this, we created lagged features, utilized rolling trends, and filtered for metros with more complete records. While these steps improved data quality, they



also made the modeling process more time-consuming and required trial and error to balance historical data depth with manageable noise.

To enhance the model's predictive power, future studies could incorporate additional economic indicators that directly influence housing prices, such as mortgage rates, unemployment rates, and local economic growth metrics. Including these factors would potentially improve the robustness of predictions, especially in fluctuating economic environments. Additionally, expanding the geographical scope to include multiple cities could offer insights into the transferability to our mode. Analyzing whether similar trends hold in cities with varying economic conditions would validate the model's broader applicability.

Another potential route for future research involved employing ensemble methods to combine linear and tree-based models. Hybrid approaches that balance linear trend detection with non-linear pattern recognition may mitigate the weaknesses identified in the tree-based models.

Our findings suggest that when venturing to forecast short-term home value trends, individuals should prioritize regularized linear models when recent historical data shows stable trends. Stakeholders may find that including overly complex or non-linear models can inadvertently introduce noise without significantly enhancing accuracy. By carefully selecting predictive features, such as recent price trends, analysts can achieve reliable forecasts without unnecessary computational complexity.

In conclusion, the study demonstrates the potential for leveraging historical home value data in forecasting future prices while highlighting the importance of model simplicity when working with momentum-driven datasets. The results offer a practical foundation for real estate

forecasting and encourage further exploration into integrating broader economic variables and hybrid modeling techniques.

## References

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